

Chaehyeong Lee

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Research Interests

Geofluid Dynamics: Cross-scale upper-ocean mixing processes; Mesoscale eddies.

Physical Oceanography: Long-term change in the upper-ocean thermal state; Ocean's role in climate systems.

Education

University of Colorado Boulder

Ph.D. in Atmospheric and Oceanic Sciences

Advisor: Dr. Donata Giglio & Co-Advisor: Dr. Aneesh Subramanian

Boulder, CO, USA

Aug. 2024 – present

Yonsei University

M.S. in Atmospheric Sciences

Advisor: Dr. Hajoong Song

Seoul, Rep. of Korea

Aug. 2023

Yonsei University

B.S. in Atmospheric Sciences

Advisor: Dr. Hajoong Song

Seoul, Rep. of Korea

Feb. 2022

Publications

Under Review.....

1. Lee, C., Giglio, D., Subramanian, A. C., Han, W., Capotondi, A., Du, D., Molod, A., & Ruiz-Xomochuk, V. (2026). Effects of Satellite Sea Surface Salinity Assimilation on the Representation of Upper-Ocean Thermohaline Structure and Variability in the NASA GEOS-S2S-2 model. *Quarterly Journal of the Royal Meteorological Society*

In Preparation.....

1. Lee, C., Giglio, D., Subramanian, A. C., Mazloff, M., & Arbic, B. K.

Published.....

1. Lee, C., Song, H., Choi, Y., Cho, A., & Marshall, J. (2025). Observed Multi-decadal Increase in the Surface Ocean's Thermal Inertia. *Nature Climate Change*, **15**, 308-314. doi:10.1038/s41558-025-02245-w

Research Experience

Dr. Giglio's Research Group, CU Boulder

Research Assistant

Boulder, CO, USA

Aug. 2024 – present

Climate Processes and Predictability Group, CU Boulder

Research Assistant

Boulder, CO, USA

Aug. 2024 – present

Air-Sea Modeling Lab, Yonsei University

Research Assistant

Seoul, Rep. of Korea

Dec. 2020 – Aug. 2024

Teaching Experience

Data Science Laboratory <i>Teaching Assistant</i>	Boulder, CO, USA <i>Spring 2026</i>
Climate & Civilization <i>Teaching Assistant</i>	Seoul, Rep. of Korea <i>Spring 2023</i>
Physical Oceanography <i>Teaching Assistant</i>	Seoul, Rep. of Korea <i>Fall 2022</i>

Awards & Scholarships

2025: ATOC Fellowship, Department of Atmospheric and Oceanic Sciences, University of Colorado Boulder
2025: Academic Research Grants (GCP research credits), Google LLC
2024: Outstanding Thesis Award, Yonsei University Graduate School, Yonsei University
2022–2023: Full tuition merit scholarship (for the top 2 graduate students), Yonsei University
2022: High Honors for academic performance, Yonsei University
2020–2021: Jilli Scholarship for academic performance, Yonsei University

Patent

Song, H., & Lee, C. (2025). *Evaluation system and method of persistence of SST anomalies using autocorrelation coefficient and arctangent regressive model*. Rep. of Korea Patent #1028135790000. doi:10.8080/1020220157159

Invited Talk

NASA Salinity Telecon <i>Assessing the Impact of Satellite Sea Surface Salinity Assimilation on Vertical Structure of the Upper Ocean in the NASA GEOS-S2S 2.</i>	Virtual Meeting <i>Dec. 2025</i>
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Selected Presentations

2027 American Meteorological Society Annual Meeting (poster) <i>Ocean Advection Drives Mixed Layer Temperature Variance at 10-100 km in the Gulf Stream Extension.</i> <i>Lee, C., Giglio, D., & Subramanian, A. C.</i>	Denver, CO, USA <i>Jan. 2027 (scheduled)</i>
US Climate Variability and Predictability (poster) <i>Enhanced Upper-Ocean Horizontal Advection in a High-Resolution Climate Model Improves the Representation of High-Frequency Near-Surface Temperature Variability.</i> <i>Lee, C., Giglio, D., & Subramanian, A. C.</i>	Boulder, CO, USA <i>Sep. 2026</i>
Ocean Science Meeting 2026 (poster) <i>Assessing the Impact of Satellite Sea Surface Salinity Assimilation on the Upper Ocean Thermal State in the NASA GEOS S2S-v2 Model.</i> <i>Lee, C., Giglio, D., Subramanian, A. C., Han, W., Capotondi, A., Du, D., & Molod, A.</i>	Glasgow, Scotland <i>Feb. 2026</i>

American Geophysical Union Fall Meeting (poster) <i>The Increasing Trend of Persistence of Sea Surface Temperature in the Past 40 Years.</i> <i>Lee, C., Song, H., Cho, A., & Tak, Y.</i>	Chicago, IL, USA <i>Dec. 2022</i>
Korean Society of Oceanography Spring Conference (talk) <i>Increasing Persistence of SST Anomalies and Duration of Marine Heatwaves.</i> <i>Lee, C., Song, H., Cho, A., & Tak, Y.</i>	Jeju, Rep. of Korea <i>Jun. 2022</i>

Workshops

Broadening Earth System Research Across Scales with High-Resolution Modeling Workshop <i>US Climate Variability and Predictability</i>	Boulder, CO, USA <i>Sep. 2026</i>
Data Assimilation Training and Science Workshop <i>Consortium for Advanced Data Assimilation Research and Education & NOAA EPIC</i>	Norman, OK, USA <i>Jun. 2026</i>
14th International Workshop on Modeling the Ocean 2024 <i>International Workshop on Modeling the Ocean</i>	Sapporo, Japan <i>Jun. 2024</i>
User Training for the Glosea 6 Climate Prediction Model <i>Korea Meteorological Administration</i>	Jeju, Rep. of Korea <i>Jan. 2022</i>
Deep Learning Training: Fundamentals of Deep Learning <i>NVIDIA Deep Learning Institute</i>	Gonju, Rep. of Korea <i>Jan. 2022</i>

Academic Service

Peer Reviewer: *Journal of Climate*

Technical Skills

Programming: Python (xarray, dask, Pangeo), Julia
Modeling: MITgcm, Oceananigans
HPC: Parallel/distributed computing, NCAR Casper/Derecho clusters
Tools: Git, Linux shell scripting, LaTeX